Software Test Plan (STP) - TaskBoard

Mini Project Management Tool (Jira-style web app)

Project	TaskBoard: Mini Project Management Tool
Version	1.0
Authors	Member 3 (Tester / QA), with input from Members 1, 2 and 4
Date	28-10-2026
Status	Draft for Sprint 1 and Sprint 2 execution

1. Introduction

Purpose: This document defines the test plan for TaskBoard v1.0. It sets out objectives, scope, strategy, resources, schedule and responsibilities for testing across Sprint 1 (27-Oct to 31-Oct-2026) and Sprint 2 (02-Nov to 07-Nov-2026).

Scope: Testing covers user registration and login, projects, issues (create, edit, delete, assign, status change), the Kanban board, comments, sprints (create, start, complete, summary) and sprint/backlog filtering. It also covers the non-functional requirements for usability, performance, security, maintainability and portability. Notifications, file attachments, drag and drop, real-time sync, fine-grained permissions and email integration are out of scope for the product, so they are out of scope for testing.

References: TaskBoard SRS (functional requirements FR-01 to FR-13, non-functional requirements NFR-01 to NFR-05), TaskBoard SAD, Team Playbook (REST API in section 4.4, manual test cases in section 12), tests/test_api.py, GitHub Actions CI workflow.

Definitions: SRS (Software Requirements Specification), SAD (System Architecture Document), RTM (Requirements Traceability Matrix), CI (Continuous Integration), CD (Continuous Delivery), TC (Test Case), FR/NFR (Functional / Non-Functional Requirement), XSS (Cross-Site Scripting), QA (Quality Assurance).

2. Test Items

- Authentication module (register, login, logout, current user, user list)
- Projects module (create, list, delete, unique project key)
- Issues module (create with auto key, list with filters, get, edit, delete, validation)
- Comments module
- Sprints module (create, start, complete, summary)
- Frontend UI: login/register screens, project selector, Kanban board, create-issue panel, sprint toolbar and filter
- Build and delivery pipeline: GitHub Actions CI (pytest on every PR) and CD (Docker image on merge to main)

3. Features to be Tested

Features are mapped to SRS requirement IDs.

ID	Requirement	Priority
FR-01	Register with name, email and password	Must
FR-02	Log in and log out	Must
FR-03	Create and list projects with a unique short key	Must
FR-04	Create issues with type, priority, story points and auto-generated key (TB-1, TB-2)	Must
FR-05	Kanban board grouped by status	Must
FR-06	Change an issue's status	Must
FR-07	Edit and delete issues	Should

ID	Requirement	Priority
FR-08	Assign an issue to a team member	Must
FR-09	Comment on an issue	Should
FR-10	Create sprints and put issues into them	Must
FR-11	Start and complete a sprint; unfinished issues return to the backlog	Should
FR-12	Sprint summary of total vs completed story points	Could
FR-13	Filter the board by sprint or backlog	Should
NFR-01	Usability: new user creates a project and an issue within 2 minutes	-
NFR-02	Performance: API responses under 500 ms for up to 1000 issues	-
NFR-03	Security: hashed passwords, server-side validation, XSS-safe rendering	-
NFR-04	Maintainability: PEP 8 style, automated tests for core API behaviour	-
NFR-05	Portability: Windows, macOS, Linux with Python 3.10+	-

4. Features Not to be Tested

- Flask, SQLAlchemy, SQLite and Werkzeug internals (third-party libraries, assumed tested by their maintainers)
- Out-of-scope features: notifications, attachments, drag and drop, real-time sync, role-based permissions, email integration
- Cross-browser pixel-perfect layout (only latest Chrome and one other browser are checked for function)
- Production hosting, TLS termination and load balancing (the app is run locally or from a Docker image)

5. Test Approach / Strategy

Levels:

- **Unit / API tests:** pytest tests in tests/test_api.py exercise every endpoint through Flask's test client against an in-memory SQLite database
- **Integration tests:** UI to API to database flows (for example create issue in the UI and verify it persists after reload)
- **System tests:** end-to-end manual pass of all test cases on a clean clone
- **Acceptance tests:** the team verifies each story's acceptance criteria (SRS section 3.3) before a ticket moves to Done

Types:

- Functional testing of all FR items
- Regression testing: full automated run on every PR plus a manual regression pass at the end of each sprint
- Performance testing of the issue list endpoint with 1000 seeded issues (NFR-02)
- Usability testing: a teammate who has not used the app creates a project and an issue, timed (NFR-01)
- Portability check: run from a clean clone on at least two operating systems (NFR-05)

Entry criteria: Feature branch merged into develop through a reviewed PR, CI green, test data and environment ready, Jira ticket in In Review.

Exit criteria: 100% of planned test cases executed, all automated tests pass in CI, 0 critical and 0 major defects open, all Must-priority requirements verified, all story acceptance criteria satisfied, and the final test report is produced.

5.1 Security Validation

- Passwords stored as hashes only (check the database; no plain text)
- Every endpoint except register, login and logout returns 401 when no session exists

- Server-side validation rejects invalid type, status, priority, story points, assignee, sprint and date inputs with HTTP 400
- Only the project owner can delete a project (HTTP 403 for others)
- XSS: an issue title containing a script tag must render as plain text (TC-20)
- SQL injection: inputs such as ' OR 1=1 -- in login and filters must not change behaviour (ORM parameterised queries)
- Session cookie is cleared on logout and protected pages are inaccessible afterwards

6. Test Environment

Item	Details
Hardware	Team laptops (Windows, macOS or Linux), 8 GB RAM recommended
Software	Python 3.10+, Flask 3.0.3, Flask-SQLAlchemy 3.1.1, SQLite, latest Chrome plus one other browser
Test tools	pytest 8.3.2 (automated API tests), GitHub Actions (CI and CD), Postman or curl (manual API checks), browser DevTools (UI and XSS checks), Jira (defect tracking), Docker (release image check)
Test data	Dummy users (for example alice@test.com, bob@test.com), project key TB, issues of every type and priority, sprints with 3 and 5 point issues, a script that seeds 1000 issues for the performance check
Environments	Local (each member), CI runner (ubuntu-latest), clean-clone environment used for the final regression

7. Test Schedule

Milestone	Date	Owner
Test plan submitted and reviewed against backlog	27-Oct-2026 to 28-Oct-2026	Member 3
Auth and project tests written and run (FR-01 to FR-03)	28-Oct-2026	Member 3
Issue and comment tests, first manual pass (FR-04 to FR-09)	29-Oct-2026	Member 3
Integration, bug retest, Sprint 1 regression	30-Oct-2026	Member 3
Sprint 1 test summary, v0.1 tag	31-Oct-2026	Members 3 and 4
Buffer day if Sprint 1 spills over	01-Nov-2026	All
Sprint test cases written (FR-10 to FR-13)	02-Nov-2026	Member 3
Full regression run, bugs raised	03-Nov-2026 to 04-Nov-2026	Member 3
Code freeze for new features, bug fixes only; update SRS, SAD, test plan	05-Nov-2026	Members 3 and 4
Final regression and final test report	06-Nov-2026	Member 3
Merge to main, tag v1.0, freeze	07-Nov-2026	Member 4

8. Test Deliverables

- Test Plan (this document)
- Test cases (manual table TC-01 to TC-27 and automated tests)
- Test scripts: tests/test_api.py and the performance seed script
- Test data
- Test execution logs (pytest output, CI run links, manual result sheet)
- Defect reports (Jira bug tickets using the team bug template)
- Test Summary Report (per sprint and final)
- docs/deviations.md for any change from this plan

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead / Tester (Member 3)	<Name>	Prepare plan, write and execute test cases, own tests/test_api.py, log and verify bugs, produce test reports
Backend Developer (Member 1)	<Name>	Fix backend defects, write unit tests for own code, support triage
Frontend Developer (Member 2)	<Name>	Fix UI defects, support UI test execution and demo checks
DevOps / Repo Manager (Member 4)	<Name>	Keep CI/CD green, merge only after QA verification, maintain Jira board and burndown

There is no Product Owner. Acceptance and sign-off are decided jointly by the team in sprint review.

10. Risks and Mitigation

Risk	Mitigation
Short schedule (two sprints) leaves little time for testing	Write tests alongside features; run CI on every PR; use the buffer day on 01-Nov
Late merges cause untested code on develop	Branch protection requires a PR, one approval and passing CI; QA verifies before Done
Frontend and backend drift apart (API contract changes)	Treat the REST API table in the SRS as the contract; API tests catch breaking changes
Environment differences between team machines	Pin versions in requirements.txt; run the final regression from a clean clone and Docker image
Single tester is a bottleneck	Developers write unit tests for their own code; all members run the manual cases once before release
Late bugs after code freeze	Freeze new features on 05-Nov; only bug fixes allowed; final regression on 06-Nov

11. Assumptions and Dependencies

- Backend API follows the documented endpoints and status codes
- SQLite and an in-memory test database are available on all machines and in CI
- GitHub Actions runners are available; CI runs pytest on every PR and push
- Test users and projects can be created freely since there is no external system
- Jira and GitHub accounts are available to all four members

12. Suspension and Resumption Criteria

Suspend testing if:

- The application fails to start or CI is red on develop for more than 4 hours
- A blocking defect (for example login broken) stops more than 30% of test cases
- Test data or database is corrupted and cannot be reset

Resume testing when:

- The blocking defect is fixed, merged and verified by a smoke test
- The environment is rebuilt and CI is green again

13. Test Case Management and Traceability

An RTM maps SRS requirements to test cases. TC-01 to TC-20 come from the team playbook (section 12). TC-21 to TC-27 are added by this plan to cover requirements that had no manual case. Automated tests are listed by function name from tests/test_api.py.

13.1 Requirements Traceability Matrix

Requirement	Manual test cases	Automated tests
FR-01 Register	TC-01, TC-02, TC-03	test_register_and_me, test_duplicate_email_rejected, test_short_password_rejected
FR-02 Login / logout	TC-04, TC-05, TC-06	test_login_wrong_password, test_login_success_after_logout, test_requires_login
FR-03 Projects	TC-07, TC-08	test_project_key_validation, test_duplicate_project_key
FR-04 Create issue	TC-09, TC-10, TC-11	test_issue_keys_increment, test_issue_requires_title, test_invalid_story_points
FR-05 Kanban board	TC-09, TC-12, TC-23	(UI only)
FR-06 Change status	TC-12	test_status_update, test_invalid_status_rejected
FR-07 Edit / delete issue	TC-13, TC-24	test_delete_issue
FR-08 Assign issue	TC-14	test_assign_issue
FR-09 Comments	TC-21, TC-22	test_comment_and_empty_comment
FR-10 Create sprint	TC-15	test_invalid_sprint_dates
FR-11 Start / complete sprint	TC-16, TC-17	test_only_one_active_sprint, test_complete_sprint_returns_unfinished_to_backlog
FR-12 Sprint summary	TC-18	test_sprint_summary_points
FR-13 Sprint / backlog filter	TC-15, TC-19	(UI only)
NFR-01 Usability	TC-25	-
NFR-02 Performance	TC-26	-
NFR-03 Security	TC-20, TC-06	test_requires_login
NFR-04 Maintainability	Run flake8 or pycodestyle on app/ and tests/	CI runs pytest on every PR
NFR-05 Portability	TC-27	CI on ubuntu-latest

13.2 Manual Test Case Catalogue

ID	Req.	Steps	Expected result
TC-01	FR-01	Register with valid name, email, password	Account created, app opens
TC-02	FR-01	Register with the same email again	Error: email already registered
TC-03	FR-01	Register with a 3-character password	Error shown, no account created
TC-04	FR-02	Log in with correct credentials	App opens, name shown in header
TC-05	FR-02	Log in with a wrong password	Error: invalid email or password
TC-06	FR-02	Log out, then reload the page	Login screen is shown
TC-07	FR-03	Create project with key TB and a name	Project appears in the dropdown
TC-08	FR-03	Create project with key 1	Error about the key
TC-09	FR-04	Add issue with a title only	Card TB-1 appears in To Do
TC-10	FR-04	Add issue with an empty title	Error: title is required
TC-11	FR-04	Add a second issue	Key is TB-2

ID	Req.	Steps	Expected result
TC-12	FR-06	Change a card's status to In Progress	Card moves to the In Progress column
TC-13	FR-07	Delete an issue and confirm	Card disappears
TC-14	FR-08	Create an issue assigned to a teammate	Assignee name shows on the card
TC-15	FR-10	Create a sprint, add an issue while it is selected	Issue appears only under that sprint filter
TC-16	FR-11	Start Sprint 1, then try to start Sprint 2	Error: another sprint is already active
TC-17	FR-11	Complete the active sprint with one Done and one To Do issue	To Do issue moves to Backlog, Done issue stays
TC-18	FR-12	Select a sprint with 3 and 5 point issues, 3 done	Shows 3 / 8 points done
TC-19	FR-13	Select Backlog in the filter	Only issues without a sprint are shown
TC-20	NFR-03	Create an issue titled with an HTML script tag	Shown as plain text, nothing executes
TC-21	FR-09	Open an issue and add a comment	Comment appears with author and time
TC-22	FR-09	Submit an empty comment	Error: comment cannot be empty
TC-23	FR-05	Create issues in different statuses and view the board	Cards are grouped in the correct status columns
TC-24	FR-07	Edit an issue's title, priority and story points	Changes are saved and shown after reload
TC-25	NFR-01	New user registers, creates a project and an issue (timed)	Completed in under 2 minutes
TC-26	NFR-02	Seed 1000 issues and call the issue list endpoint	Response under 500 ms
TC-27	NFR-05	Run the app from a clean clone on two operating systems	App starts and TC-01 to TC-11 pass

For every cycle, record the actual result and Pass or Fail against each case and link every failure to a Jira bug ticket.

14. Test Metrics and Reporting

Metrics collected:

- Percentage of test cases executed
- Percentage passed and failed
- Number of automated tests passing in CI
- Defect density (bugs per feature area) and defects by severity (Critical, Major, Minor)
- Defect aging (days from raised to verified closed)
- Requirement coverage (requirements with at least one executed test)

Reports:

- Daily execution status in the stand-up
- Sprint 1 test summary (31-Oct-2026) and Sprint 2 test summary (07-Nov-2026)
- Final Test Summary Report with executed, passed, failed and open-bug counts

15. Approvals

Role	Name	Signature / Date
QA Lead (Member 3)		
Backend Developer (Member 1)		
Frontend Developer (Member 2)		
DevOps / Repo Manager (Member 4)		